

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 15

Section 1 : MCQ

1. What will be the output of the following code?

```
class Demo {  
    void printMessage() {  
        System.out.println("Hello from Demo");  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        Demo d = new Demo();  
        d.printMessage();  
    }  
}
```

Answer

Hello from Demo

Status : Correct

Marks : 1/1

2. What will be the output of the following code?

```
class Box {  
    int volume(int l, int b, int h) {  
        return l * b * h;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Box b = new Box();  
        System.out.println(b.volume(2, 3, 4));  
    }  
}
```

Answer

24

Status : Correct

Marks : 1/1

3. What is the output of the following code?

```
class Box {  
    int height;  
    Box(int height) {  
        this.height = height;  
    }  
    void modifyHeight(Box b) {  
        b.height += 10;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Box b1 = new Box(20);  
        b1.modifyHeight(b1);  
    }  
}
```

```
        System.out.println(b1.height);  
    }  
}
```

Answer

30

Status : Correct

Marks : 1/1

4. What will be the output of the following code?

```
class Test {  
    private int value;  
    Test(int value) {  
        this.value = value;  
    }  
    public int getValue() {  
        return value;  
    }  
}  
public class Main {  
    public static void main(String[] args) {  
        Test obj = new Test(10);  
        System.out.println(obj.value);  
    }  
}
```

Answer

Compile-time error

Status : Correct

Marks : 1/1

5. What will be the output of the following code?

```
class Box {  
    int length = 5;  
    int width = 4;  
    int area() {
```

```
        return length * width;
    }

    public static void main(String[] args) {
        Box b = new Box();
        System.out.println("Area = " + b.area());
    }
}
```

Answer

Area = 20

Status : Correct

Marks : 1/1

6. What will be the output of the following code?

```
class Alpha {
    void greet(String name) {
        System.out.println("Hello " + name);
    }
}

public class Main {
    public static void main(String[] args) {
        Alpha obj = new Alpha();
        obj.greet("Anu");
    }
}
```

Answer

Hello Anu

Status : Correct

Marks : 1/1

7. What will be the output of the following code?

```
class A {
    int y = 30;
}
```

```
public class Main {  
    public static void main(String[] args) {  
        A a1 = new A();  
        A a2 = new A();  
        a1.y = 50;  
        System.out.println(a2.y);  
    }  
}
```

Answer

30

Status : Correct

Marks : 1/1

8. What will be the output of the following code?

```
class A {  
    int val = 20;  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        A obj1 = new A();  
        A obj2 = obj1;  
        obj2.val += 5;  
        System.out.println(obj1.val);  
    }  
}
```

Answer

25

Status : Correct

Marks : 1/1

9. What will be the output of the following code?

```
class Person {  
    int age = 18;
```

```
}  
public class Main {  
    public static void main(String[] args) {  
        Person p = new Person();  
        p.age += 2;  
        System.out.println("Age: " + p.age);  
    }  
}
```

Answer

Age: 20

Status : Correct

Marks : 1/1

10. What will be the output of the following code?

```
class Person {  
    String name;  
    void setName(String n) {  
        name = n;  
    }  
    void printName() {  
        System.out.println(name);  
    }  
}  
  
class Test {  
    public static void main(String[] args) {  
        Person p = new Person();  
        p.printName();  
    }  
}
```

Answer

null

Status : Correct

Marks : 1/1

11. What will be the output of the following code?

```
class A {  
    int x = 50;  
}  
  
public class Main {  
    public static void main(String[] args) {  
        A obj1 = new A();  
        A obj2 = obj1;  
        obj2.x = 100;  
        System.out.println(obj1.x);  
    }  
}
```

Answer

100

Status : Correct

Marks : 1/1

12. What will be the output of the following code?

```
class MathUtils {  
    int add(int x) {  
        return x + x;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        MathUtils m = new MathUtils();  
        System.out.println(m.add(5));  
    }  
}
```

Answer

10

Status : Correct

Marks : 1/1

13. What will be the output of the following code?

```
class Sample {  
    int x = 10;  
  
    void display() {  
        System.out.println("x = " + x);  
    }  
  
    public static void main(String[] args) {  
        Sample s = new Sample();  
        s.display();  
    }  
}
```

Answer

x = 10

Status : Correct

Marks : 1/1

14. What will be the output of the following code?

```
class Ball {  
    int size = 11;  
}  
  
class Game {  
    public static void main(String[] args) {  
        Ball b1 = new Ball();  
        Ball b2 = new Ball();  
        b2.size = 10;  
        System.out.println(b1.size);  
    }  
}
```

Answer

11

Status : Correct

Marks : 1/1

15. What will be the output of the following code?

```
class A {  
    int p = 5;  
    int q = 2;  
}  
  
class Main {  
    public static void main(String[] args) {  
        A obj = new A();  
        System.out.println(obj.p + obj.q);  
    }  
}
```

Answer

7

Status : Correct

Marks : 1/1

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_Q2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

You are working as a developer for CityBank, which wants to build a basic account management system.

Each customer at the bank has:

An Account Number (integer) A Customer Name (string) An Initial Balance (double)

The bank allows two types of transactions:

Deposit – increases the balance. Withdrawal – decreases the balance only if enough funds are available.

If the withdrawal amount is greater than the balance, the withdrawal should not happen, and the balance should remain the same.

You are required to implement this system using:

A class with attributes for account details. A constructor to initialize account details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customers.

Finally, display each customer's account details after all transactions.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the account number (integer).
- The following line contains the customer name (string).
- The next line contains the initial balance (double).
- The next line contains the deposit amount (double).
- The next line contains the withdrawal amount (double).

Output Format

For each customer, print the details in the following format:

1. Account Number: <account_number>
2. Customer Name: <customer_name>
3. Final Balance: <final_balance> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Rahul Sharma

5000

2000

3000

Output: Account Number: 1234

Customer Name: Rahul Sharma

Final Balance: 4000.0

Answer

```
// You are using Java
import java.util.*;
class Bank{
    private int acc;
    private String cust;
    private double bal;

    public Bank(int acc,String cust,double bal){
        this.acc=acc;
        this.cust=cust;
        this.bal=bal;
    }
    public int getAcc(){
        return acc;
    }
    public String getName(){
        return cust;
    }
    public double getbal(){
        return bal;
    }
    public void setAcc(int acc){
        this.acc=acc;
    }
    public void setName(String cust){
        this.cust=cust;
    }
    public void deposit(double amnt){
        if (amnt>=0){
            bal+=amnt;
        }
    }
    public void withdraw(double amnt){
        if (amnt<=bal){
            bal-=amnt;
        }
    }
}
class main{
```

```
public static void main(String args[]){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    sc.nextLine();
    for(int i=0;i<n;i++){
        int acc=sc.nextInt();
        sc.nextLine();
        String cust=sc.nextLine();
        double bal=sc.nextDouble();
        double dep=sc.nextDouble();
        double wdr=sc.nextDouble();
        Bank acc1=new Bank(acc,cust,bal);
        acc1.deposit(dep);
        acc1.withdraw(wdr);
        System.out.printf("Account Number: %d\n",acc1.getAcc());
        System.out.printf("Customer Name: %s\n",acc1.getName());
        System.out.printf("Final Balance: %.1f",acc1.getbal());
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Neha is working as a developer for CityElectricity Board, which wants to build a household electricity billing system.

Each customer's electricity account has:

A Customer ID (integer) A Customer Name (string) Units Consumed (double)

The electricity bill is calculated based on these rules:

For the first 100 units 5 units charge per unit For the next 100 units (101–200) 7 units charge per unit For units above 200 10 units charge per unit If the total bill exceeds 2000 units, a 5% discount is applied on the final bill.

Neha has been asked to implement this system using:

A class with attributes for customer details. A constructor to initialize customer details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customers.

Finally, display each customer's details and final bill amount.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the Customer ID (integer).
- The following line contains the Customer Name (string).
- The next line contains the Units Consumed (double).

Output Format

For each customer, print the details in the following format:

Customer ID: <customer_id>

Customer Name: <customer_name>

Final Bill: <final_bill> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

80

Output: Customer ID: 1001

Customer Name: Ravi Kumar

Final Bill: 400.0

Answer

// You are using Java

```
import java.util.*;
class ElectricityBill{
    private int CusId;
    private String Name;
    private double Units;
    public ElectricityBill(int CusId,String Name,double Units){
        this.CusId=CusId;
        this.Name=Name;
        this.Units=Units;
    }
    public void SetId(int Id){
        CusId=Id;
    }
    public void SetName(String name){
        Name=name;
    }
    public void SetUnits(double units){
        Units=units;
    }
    public int GetId(){
        return CusId;
    }
    public String GetName(){
        return Name;
    }
}
class main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        for (int i=0;i<n;i++){
            int id=sc.nextInt();
            sc.nextLine();
            String name=sc.nextLine();
            double units=sc.nextDouble();
            double bill;
            if (units<=100){
                bill=units*5.0;
            }
            else if(units>100 && units<=200){
                bill=500+((units-100)*7);
            }
        }
    }
}
```

```
}
else {
    bill=1200+((units-200)*10);
}
if(bill>2000){
    bill=bill-(bill*0.05);
}
ElectricityBill EB=new ElectricityBill(id,name,units);
System.out.printf("Customer Id: %d\n",EB.GetId());
System.out.printf("Customer Name: %s\n",EB.GetName());
System.out.printf("Final Bill: %.1f\n",bill);
}
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

You are working as a developer for CityCab, a taxi service company that wants to build a ride fare management system.

Each customer booking has:

A Booking ID (integer) A Customer Name (string) A Distance Travelled in km (double)

The fare calculation rules are:

Base Fare = 50 units (flat charge for every ride). Per km charge = 10 units/km. If the distance is greater than 20 km, a 10% discount is applied on the total fare.

You are required to implement this system using:

A class with attributes for booking details. A constructor to initialize booking details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customer rides.

Finally, display each booking's details and final fare.

Input Format

The first line of input contains an integer N, representing the number of bookings.

For each booking:

- The next line contains the booking ID (integer).
- The following line contains the customer's name (string).
- The next line contains the distance travelled (double).

Output Format

For each booking, print the details in the following format:

1. Booking ID: <booking_id>
2. Customer Name: <customer_name>
3. Final Fare: <final_fare> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
1234
Rahul Sharma
15

Output: Booking ID: 1234
Customer Name: Rahul Sharma
Final Fare: 200.0

Answer

```
// You are using Java
import java.util.Scanner;
```

```
class Booking {
    private int bookingId;
    private String customerName;
    private double distance;
    private double fare;

    public Booking(int bookingId, String customerName, double distance) {
        this.bookingId = bookingId;
        this.customerName = customerName;
        this.distance = distance;
        calculateFare();
    }

    public void setBookingId(int bookingId) {
        this.bookingId = bookingId;
    }

    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }

    public void setDistance(double distance) {
        this.distance = distance;
        calculateFare();
    }

    public int getBookingId() {
        return bookingId;
    }

    public String getCustomerName() {
        return customerName;
    }

    public double getDistance() {
        return distance;
    }

    public double getFare() {
        return fare;
    }
}
```

```

private void calculateFare() {
    fare = 50 + distance * 10;
    if (distance > 20) {
        fare = fare - (fare * 0.1);
    }
}

}

class CityCabApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine());
        for (int i = 0; i < n; i++) {
            int id = Integer.parseInt(sc.nextLine());
            String name = sc.nextLine();
            double distance = Double.parseDouble(sc.nextLine());
            Booking booking = new Booking(id, name, distance);
            System.out.println("Booking ID: " + booking.getBookingId());
            System.out.println("Customer Name: " + booking.getCustomerName());
            System.out.printf("Final Fare: %.1f\n", booking.getFare());
        }
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_Q5

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Ram is working as a developer for BrightEdu Coaching Center, which wants to build a student fee management system.

Each student's enrollment has:

An Enrollment ID (integer) A Student Name (string) The Number of Subjects (integer)

The fee calculation rules are:

Registration Fee = 1000 units (flat for every student). Per Subject Fee = 800 units. If the student enrolls in more than 5 subjects, a 20% scholarship (discount) is applied on the total fee.

Ram has been asked to implement this system using:

A class with attributes for student details. A constructor to initialize student details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent student enrollments.

Finally, display each student's details and final fee.

Input Format

The first line of input contains an integer N, representing the number of students.

For each student:

- The next line contains the Enrollment ID (integer).
- The following line contains the student's name (string).
- The next line contains the Number of subjects (integer).

Output Format

For each student, print the details in the following format:

- Enrollment ID: <enrollment_id>
- Student Name: <student_name>
- Final Fee: <final_fee> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Ravi Kumar

3

Output: Enrollment ID: 1234

Student Name: Ravi Kumar

Final Fee: 3400.0

Answer

```
// You are using Java
```

```
import java.util.*;
```

```
class Enroll{
```

```
private int Id;
private String name;
private int num;

public Enroll(int Id,String name,int num){
    this.Id=Id;
    this.name=name;
    this.num=num;
}
public void setId(int Id){
    this.Id=Id;
}
public void setName(String name){
    this.name=name;
}
public void setNum(int num){
    this.num=num;
}
public int getId(){
    return Id;
}
public String getName(){
    return name;
}
public int getNum(){
    return num;
}
public double calFare(){
    double fee=1000+(num*800);
    if (num>5){
        fee=fee-(fee*0.2);
    }
    return fee;
}
}

public class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        for (int i=0;i<n;i++){
            int Id=sc.nextInt();
```

```
        sc.nextLine();
        String name=sc.nextLine();
        int num=sc.nextInt();
        Enroll E1=new Enroll(Id,name,num);
        System.out.printf("Enrollment ID: %d\n",E1.getId());
        System.out.printf("Student Name: %s\n",E1.getName());
        System.out.printf("Final Fee: %.1f",E1.calFare());
    }
}
```

Status : Correct

Marks : 10/10

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 5_PAH

Attempt : 1
Total Mark : 50
Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Neha is working as a developer for CityMovie Theatre, which wants to build a system to calculate total ticket cost for movie-goers based on the number of tickets and type of seats booked.

Each customer's booking has:

Booking ID (integer) Customer Name (string) Number of Tickets (integer) Seat Type (string: "Standard", "Premium", "VIP")

The ticket prices are:

Standard – 250 units per ticket Premium – 400 units per ticket VIP – 600 units per ticket

The calculation rules:

Total Amount = Number of Tickets × Seat Price

If a customer books more than 4 tickets, they get a 10% discount on the total amount.

If the booking is for VIP seats and the total amount exceeds 3000 units, a 5% luxury tax is added after any discount.

Neha has been asked to implement this system using:

A class with attributes for booking details. A constructor to initialize booking details. Getter and Setter methods to retrieve and update booking details if required. A method to calculate the final ticket cost. Objects of the class to represent bookings.

Finally, display each customer's details and final ticket amount.

Input Format

The first line contains an integer N, representing the number of bookings.

For each booking:

- The next line contains the Booking ID (integer).
- The next line contains the Customer Name (string).
- The next line contains Number of Tickets (integer).
- The next line contains Seat Type ("Standard", "Premium", or "VIP").

Output Format

For each booking, print:

- Booking ID: <booking_id>
- Customer Name: <customer_name>
- Final Ticket Amount: <final_amount> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

3

Standard

Output: Booking ID: 1001

Customer Name: Ravi Kumar

Final Ticket Amount: 750.0

Answer

// You are using Java

import java.util.*;

class Ticket{

private int Id;

private String name;

private int num;

private String type;

public Ticket(int Id,String name,int num,String type){

this.Id=Id;

this.name=name;

this.num=num;

this.type=type;

}

public String getName(){

return name;

}

public int getId(){

return Id;

}

public double calFare(){

double fare;

if (type.equals("Standard")){

fare=num*250;

}

else if (type.equals("Premium")){

fare=num*400;

}

else {

```

        fare=num*600;
    }
    if (num>4){
        fare=fare-(fare*.1);
    }
    if (fare>3000){
        fare=fare+(fare*0.05);
    }
    return fare;
}
}

```

```

class main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        for (int i=0;i<n;i++){
            int id=sc.nextInt();
            sc.nextLine();
            String name=sc.nextLine();
            int num=sc.nextInt();
            sc.nextLine();
            String type=sc.nextLine();
            Ticket T1=new Ticket(id,name,num,type);
            System.out.printf("Booking ID: %d\n",T1.getId());
            System.out.printf("Customer Name: %s\n",T1.getName());
            System.out.printf("Final Ticket Amount: %.1f\n",T1.calFare());
        }
    }
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Each customer at the bank has an Account Number, Customer Name, and an Initial Balance. The bank allows two types of transactions:

Deposit – Increases the balance. Withdrawal – Decreases the balance, but only if enough funds are available. If the withdrawal amount exceeds the

available balance, the transaction should be skipped, and the balance should remain unchanged.

You are required to implement this banking system by:

Creating a class with the necessary attributes to store account details.

Using a constructor to initialize the account details when a new account is created. Providing setter methods to update the details if required. Providing getter methods to retrieve account details. Creating objects of this class to represent different customers, where each customer can perform deposits and withdrawals.

Instructions:

Implement the class to store account details. Implement the logic for performing deposit and withdrawal transactions. Ensure that withdrawals don't exceed the available balance. After performing the transactions, print the account number, customer name, and final balance.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the account number (integer).
- The following line contains the customer name (string).
- The next line contains the initial balance (double).
- The next line contains the deposit amount (double).
- The next line contains the withdrawal amount (double).

Output Format

For each customer, print the details in the following format:

1. Account Number: <account_number>
2. Customer Name: <customer_name>
3. Final Balance: <final_balance> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Rahul Sharma

5000

2000

3000

Output: Account Number: 1234

Customer Name: Rahul Sharma

Final Balance: 4000.0

Answer

// You are using Java

import java.util.Scanner;

class Account {

private int accountNumber;

private String customerName;

private double balance;

public Account(int accountNumber, String customerName, double balance) {

this.accountNumber = accountNumber;

this.customerName = customerName;

this.balance = balance;

}

public void setAccountNumber(int accountNumber) {

this.accountNumber = accountNumber;

}

public void setCustomerName(String customerName) {

this.customerName = customerName;

}

public void setBalance(double balance) {

this.balance = balance;

}

public int getAccountNumber() {

return accountNumber;

}

```

    public String getCustomerName() {
        return customerName;
    }

    public double getBalance() {
        return balance;
    }

    public void deposit(double amount) {
        if (amount >= 0) balance += amount;
    }

    public void withdraw(double amount) {
        if (amount <= balance) balance -= amount;
    }
}

class CityBankApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine());
        for (int i = 0; i < n; i++) {
            int accNo = Integer.parseInt(sc.nextLine());
            String name = sc.nextLine();
            double initBal = Double.parseDouble(sc.nextLine());
            double deposit = Double.parseDouble(sc.nextLine());
            double withdraw = Double.parseDouble(sc.nextLine());

            Account acc = new Account(accNo, name, initBal);
            acc.deposit(deposit);
            acc.withdraw(withdraw);

            System.out.println("Account Number: " + acc.getAccountNumber());
            System.out.println("Customer Name: " + acc.getCustomerName());
            System.out.println("Final Balance: " + acc.getBalance());
        }
        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Neha is working as a developer for CityQuiz Platform, which wants to build a system to calculate quiz scores and identify top scorers among participants.

Each participant's record has:

Participant ID (integer) Participant Name (string) An array of scores in 5 quiz rounds (integers, each between 0 and 100)

The system must calculate:

Total Score = sum of scores in all 5 rounds. Average Score = Total Score ÷ 5. If a participant scores above 80 in all rounds, a bonus of 10 points is added to the total score. Identify the Top Scorer among all participants. If two participants have the same total score, the one with the lower Participant ID is considered the top scorer.

Neha has been asked to implement this system using:

A class with attributes for participant details. A constructor to initialize participant details. Getter and setter methods to retrieve or update participant details. A method to calculate total score and average score (including bonus if applicable). Objects of the class to represent participants.

Finally, display each participant's details and announce the Top Scorer.

Input Format

The first line of input contains an integer N, representing the number of participants.

For each participant:

- Next line: Participant ID (integer)
- Next line: Participant Name (string)
- Next line: 5 integers separated by spaces (scores for 5 quiz rounds)

Output Format

For each participant:

- Participant ID: <participant_id>
- Participant Name: <participant_name>
- Total Score: <total_score>
- Average Score: <average_score>

Finally, print "Top Scorer: <participant_name> with <total_score> points"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

85 90 88 92 87

Output: Participant ID: 1001

Participant Name: Ravi Kumar

Total Score: 452

Average Score: 90

Top Scorer: Ravi Kumar with 452 points

Answer

```
import java.util.*;
```

```
class Participant {  
    private int participantId;  
    private String participantName;  
    private int[] scores;  
    private int totalScore;  
    private int averageScore;
```

```
    public Participant(int participantId, String participantName, int[] scores) {  
        this.participantId = participantId;  
        this.participantName = participantName;  
        this.scores = scores;  
        calculateScores();
```

```
}  
  
public int getParticipantId() {  
    return participantId;  
}  
  
public String getParticipantName() {  
    return participantName;  
}  
  
public int getTotalScore() {  
    return totalScore;  
}  
  
public int getAverageScore() {  
    return averageScore;  
}  
  
public void setParticipantName(String name) {  
    this.participantName = name;  
}  
  
public void setScores(int[] scores) {  
    this.scores = scores;  
    calculateScores();  
}  
  
private void calculateScores() {  
    totalScore = 0;  
    boolean bonusEligible = true;  
    for (int s : scores) {  
        totalScore += s;  
        if (s <= 80) {  
            bonusEligible = false;  
        }  
    }  
    if (bonusEligible) {  
        totalScore += 10;  
    }  
    averageScore = totalScore / scores.length;  
}
```

```

    public void display() {
        System.out.println("Participant ID: " + participantId);
        System.out.println("Participant Name: " + participantName);
        System.out.println("Total Score: " + totalScore);
        System.out.println("Average Score: " + averageScore);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = Integer.parseInt(sc.nextLine().trim());
        List<Participant> participants = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            int id = Integer.parseInt(sc.nextLine().trim());
            String name = sc.nextLine().trim();
            String[] scoreStr = sc.nextLine().split(" ");
            int[] scores = new int[5];
            for (int j = 0; j < 5; j++) {
                scores[j] = Integer.parseInt(scoreStr[j]);
            }
            participants.add(new Participant(id, name, scores));
        }

        for (Participant p : participants) {
            p.display();
        }

        Participant top = participants.get(0);
        for (Participant p : participants) {
            if (p.getTotalScore() > top.getTotalScore() ||
                (p.getTotalScore() == top.getTotalScore() && p.getParticipantId() <
                 top.getParticipantId())) {
                top = p;
            }
        }

        System.out.println("Top Scorer: " + top.getParticipantName() + " with " +
            top.getTotalScore() + " points");
    }
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Anjali is working as a developer for CityFitness Gym, which wants to build a system to calculate monthly membership fees for gym members based on the type of membership and the number of personal training sessions booked.

Each member's record has:

Member ID (integer) Member Name (string) Membership Type (string: "Basic", "Premium", "Elite") Number of Personal Training Sessions (integer)

The monthly fees are:

Basic – 1000 units Premium – 1500 units Elite – 2000 units

The cost of personal training sessions is 500 units per session.

The calculation rules:

Total Amount = Membership Fee + (Number of Personal Training Sessions × 500) If the number of sessions is more than 5, a 10% discount is applied on the total amount. If the member has Elite membership and the total amount exceeds 4000, an additional 5% service tax is added after discount.

Anjali has been asked to implement this system using:

A class with attributes for member details. A constructor to initialize member details. Getter and Setter methods to retrieve and update member details if required. A method to calculate the final monthly fee. Objects of the class to represent members.

Finally, display each member's details and the final monthly fee.

Input Format

The first line contains an integer N, representing the number of members.

For each member:

- Next line contains Member ID (integer)
- Next line contains Member Name (string)
- Next line contains Membership Type ("Basic", "Premium", "Elite")
- Next line contains Number of Personal Training Sessions (integer)

Output Format

For each member, print:

- Member ID: <member_id>
- Member Name: <member_name>
- Final Monthly Fee: <final_fee> (The final fee must be rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

Basic

3

Output: Member ID: 1001

Member Name: Ravi Kumar

Final Monthly Fee: 2500.0

Answer

// You are using Java

import java.util.*;

class member{

private int id;

private String name;

private String type;

private int num;

public member(int id,String name,String type,int num){

this.id=id;

this.name=name;

```
this.type=type;
this.num=num;
}

public int getid(){
    return id;
}

public String getname(){
    return name;
}

public String gettype(){
    return type;
}

public int getnum(){
    return num;
}

public double calFee(){
    double fee;
    if (type.equals("Basic")){
        fee=1000+(num*500);
    }
    else if (type.equals("Premium")){
        fee=1500+(num*500);
    }
    else {
        fee=2000+(num*500);
    }
    if (num>5)
        fee=fee-(fee*0.1);
    if ((type.equals("Elite"))&& (fee>4000))
        fee=fee+(fee*0.05);
    return fee;
}

}

class main{
    public static void main(String args[]){
        Scanner sc=new Scanner (System.in);
```

```

int n=sc.nextInt();
sc.nextLine();
while(n>0){
    int id=sc.nextInt();
    sc.nextLine();
    String name=sc.nextLine();
    String type=sc.nextLine();
    int num=sc.nextInt();
    member m1=new member(id,name,type,num);
    System.out.println("Member ID: "+m1.getid());
    System.out.println("Member Name: "+m1.getname());
    System.out.println("Final Monthly Fee: "+m1.calFee());
    n--;
}
}
}

```

Status : Correct

Marks : 10/10

5. Problem Statement

Ravi is working as a developer for SecureLogin Systems, which wants to build a system to evaluate the strength of user passwords.

Each user record has:

User ID (integer) User Name (string) Password (string)

The system must calculate whether a password is strong or weak.

A password is considered strong if it meets all of the following conditions:

At least 8 characters long. Contains at least one uppercase letter. Contains at least one lowercase letter. Contains at least one digit. Contains at least one special character (from !@#\$%^&*).

Ravi has been asked to implement this system using:

A class with attributes for user details. A constructor to initialize user details. Getter and setter methods to retrieve or update user details. A

method to check whether the password is strong. Objects of the class to represent users.

Finally, display each user's details and indicate whether their password is Strong or Weak.

Input Format

The first line contains an integer N, representing the number of users.

For each user:

The next line contains the User ID (integer).

The next line contains the User Name (string).

The next line contains the Password (string).

Output Format

For each user, print the details in the following format:

User ID: <user_id>

User Name: <user_name>

Password: <password>

Password Strength: <Strong/Weak>

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

Abc@1234

Output: User ID: 1001

User Name: Ravi Kumar

Password: Abc@1234

Password Strength: Strong

Answer

```
// You are using Java
import java.util.*;
```

```
class User{
    private int id;
    private String name;
    private String pass;

    public User(int id,String name,String pass){
        this.id=id;
        this.name=name;
        this.pass=pass;
    }

    public int getid(){
        return id;
    }

    public String getname(){
        return name;
    }

    public String getPass(){
        return pass;
    }

    public void check(){
        boolean dig=false,up=false,lw=false,sp=false;
        for (char ch : pass.toCharArray()){
            if(Character.isUpperCase(ch))
                up=true;
            else if (Character.isLowerCase(ch))
                lw=true;
            else if (Character.isDigit(ch))
                dig=true;
            else
                sp=true;
        }
        if ((pass.length())>=8)&&up&&lw&&dig&&sp)
```

```

        System.out.println("Strong");
    else
        System.out.println("Weak");
    }
}

class main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        sc.nextLine();
        while(n>0){
            n--;
            int id=sc.nextInt();
            sc.nextLine();
            String name=sc.nextLine();
            String pass=sc.nextLine();
            User u1=new User(id,name,pass);
            System.out.println("User ID: "+u1.getId());
            System.out.println("User Name: "+u1.getName());
            System.out.println("Password: "+u1.getPass());
            System.out.printf("Password Strength: ");
            u1.check();
        }
    }
}

```

Status : Correct

Marks : 10/10

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Scan to verify results



2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 5_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Arjun is working as a developer for CityWater Supply Board, which wants to build a household water billing system.

Each household's water account has:

A Customer ID (integer) A Customer Name (string) Liters Consumed (double)

The water bill is calculated based on these rules:

For the first 500 liters 2 per liter For the next 500 liters (501–1000) 3 per liter
For liters above 1000 5 per liter If the total bill exceeds 3000, a 10% discount is applied on the final bill.

Arjun has been asked to implement this system using:

A class with attributes for customer details. A constructor to initialize customer details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customers.

Finally, display each customer's details and final bill amount.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the Customer ID (integer).
- The following line contains the Customer Name (string).
- The next line contains the Liters Consumed (double).

Output Format

For each customer, print the details in the following format:

Customer ID: <customer_id>

Customer Name: <customer_name>

Final Bill: <final_bill> (rounded to one decimal place)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

300

Output: Customer ID: 1001

Customer Name: Ravi Kumar

Final Bill: 600.0

Answer

// You are using Java

```
import java.util.*;
```

```
class Customer {  
    private int customerId;  
    private String customerName;  
    private double litersConsumed;
```

```
    public Customer(int customerId, String customerName, double  
litersConsumed) {  
        this.customerId = customerId;  
        this.customerName = customerName;  
        this.litersConsumed = litersConsumed;  
    }
```

```
    public void setCustomerId(int customerId) {  
        this.customerId = customerId;  
    }
```

```
    public void setCustomerName(String customerName) {  
        this.customerName = customerName;  
    }
```

```
    public void setLitersConsumed(double litersConsumed) {  
        this.litersConsumed = litersConsumed;  
    }
```

```
    public int getCustomerId() {  
        return customerId;  
    }
```

```
    public String getCustomerName() {  
        return customerName;  
    }
```

```
    public double getLitersConsumed() {  
        return litersConsumed;  
    }
```

```
    public double calculateBill() {  
        double liters = this.litersConsumed;
```

```

double bill;

    if (liters <= 500) {
        bill = liters * 2;
    } else if (liters <= 1000) {
        bill = (500 * 2) + ((liters - 500) * 3);
    } else {
        bill = (500 * 2) + (500 * 3) + ((liters - 1000) * 5);
    }

    if (bill > 3000) {
        bill *= 0.9;
    }

    return bill;
}
}

```

```

class main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = Integer.parseInt(sc.nextLine().trim());
        List<Customer> customers = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            int customerId = Integer.parseInt(sc.nextLine().trim());
            String customerName = sc.nextLine().trim();
            double litersConsumed = Double.parseDouble(sc.nextLine().trim());

            Customer customer = new Customer(customerId, customerName,
litersConsumed);
            customers.add(customer);
        }

        for (Customer c : customers) {
            System.out.println("Customer ID: " + c.getCustomerId());
            System.out.println("Customer Name: " + c.getCustomerName());
            System.out.printf("Final Bill: %.1f\n", c.calculateBill());
        }

        sc.close();
    }
}

```

Status : Correct

Marks : 10/10

2. Problem Statement

Anjali is now working as a developer for the City Marathon Association, which wants to build a system to track and find the fastest runner among marathon participants.

Each runner's record has:

Runner ID (integer) Runner Name (string) An array of times (in minutes) taken in 5 marathon events (integers)

The system must calculate:

The average time of each runner (sum of all times / 5). Identify the fastest runner (the one with the lowest average time). If two or more runners have the same average time, the one with the lower Runner ID is considered the fastest runner.

Anjali has been asked to implement this system using:

A class with attributes for runner details. A constructor to initialize runner details. Getter and Setter methods to retrieve and update runner details if required. A method to calculate the average time. Objects of the class to represent runners.

Finally, display each runner's details and announce the Fastest Runner.

Input Format

The first line of input contains an integer N (number of runners).

For each runner:

- The next line contains the Runner ID (integer).
- The following line contains the Runner Name (string).
- The next line contains 5 integers separated by spaces (times in minutes for 5

marathon events).

Output Format

For each runner the output prints the following details:

- Runner ID: <runner_id>
- Runner Name: <runner_name>
- Average Time: <average_time>

Finally, print "Fastest Runner: <runner_name> with <average_time> minutes"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

240 250 245 255 260

Output: Runner ID: 1001

Runner Name: Ravi Kumar

Average Time: 250

Fastest Runner: Ravi Kumar with 250 minutes

Answer

// You are using Java

import java.util.*;

class Runner {

private int runnerId;

private String runnerName;

private int[] times;

public Runner(int runnerId, String runnerName, int[] times) {

this.runnerId = runnerId;

this.runnerName = runnerName;

this.times = times;

```
}

public void setRunnerId(int runnerId) {
    this.runnerId = runnerId;
}

public void setRunnerName(String runnerName) {
    this.runnerName = runnerName;
}

public void setTimes(int[] times) {
    this.times = times;
}

public int getRunnerId() {
    return runnerId;
}

public String getRunnerName() {
    return runnerName;
}

public int[] getTimes() {
    return times;
}

public int calculateAverageTime() {
    int sum = 0;
    for (int t : times) {
        sum += t;
    }
    return sum / times.length;
}

}

class main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        Runner[] runners = new Runner[n];
```

```

    for (int i = 0; i < n; i++) {
        int runnerId = sc.nextInt();
        sc.nextLine();
        String runnerName = sc.nextLine();

        int[] times = new int[5];
        for (int j = 0; j < 5; j++) {
            times[j] = sc.nextInt();
        }

        runners[i] = new Runner(runnerId, runnerName, times);
    }

    Runner fastestRunner = runners[0];
    int bestAvg = fastestRunner.calculateAverageTime();

    for (Runner r : runners) {
        int avg = r.calculateAverageTime();
        System.out.println("Runner ID: " + r.getRunnerId());
        System.out.println("Runner Name: " + r.getRunnerName());
        System.out.println("Average Time: " + avg);

        if (avg < bestAvg || (avg == bestAvg && r.getRunnerId() <
            fastestRunner.getRunnerId())) {
            bestAvg = avg;
            fastestRunner = r;
        }
    }

    System.out.println("Fastest Runner: " + fastestRunner.getRunnerName() + "
with " + bestAvg + " minutes");

    sc.close();
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

You are working as a developer for CityMobile, which wants to build a

basic mobile data usage management system.

Each customer has:

A Customer ID (integer) A Customer Name (string) An Initial Data Balance (in GB, double)

The company allows two types of operations:

Recharge – increases the data balance. Usage – decreases the data balance only if enough data is available.

If the usage amount is greater than the available data balance, the usage should not happen, and the balance should remain the same.

You are required to implement this system using:

A class with attributes for customer details. A constructor to initialize customer details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customers.

Finally, display each customer's details after all operations.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the Customer ID (integer).
- The following line contains the Customer Name (string).
- The next line contains the Initial Data Balance (double).
- The next line contains the Recharge Amount in GB (double).
- The next line contains the Usage Amount in GB (double).

Output Format

For each customer, print the details in the following format:

Customer ID: <customer_id>

Customer Name: <customer_name>

Final Data Balance: <final_data_balance> GB (The final balance must be rounded to one decimal place.)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1234

Ravi Kumar

5.0

2.0

3.0

Output: Customer ID: 1234

Customer Name: Ravi Kumar

Final Data Balance: 4.0 GB

Answer

// You are using Java

```
import java.util.*;
```

```
class Customer {  
    private int customerId;  
    private String customerName;  
    private double dataBalance;
```

```
    public Customer(int customerId, String customerName, double dataBalance) {  
        this.customerId = customerId;  
        this.customerName = customerName;  
        this.dataBalance = dataBalance;  
    }
```

```
    public void setCustomerId(int customerId) {  
        this.customerId = customerId;  
    }
```

```
    public void setCustomerName(String customerName) {  
        this.customerName = customerName;
```

```
}  
  
public void setDataBalance(double dataBalance) {  
    this.dataBalance = dataBalance;  
}  
  
public int getCustomerId() {  
    return customerId;  
}  
  
public String getCustomerName() {  
    return customerName;  
}  
  
public double getDataBalance() {  
    return dataBalance;  
}  
  
public void recharge(double amount) {  
    if (amount > 0) {  
        this.dataBalance += amount;  
    }  
}  
  
public void useData(double amount) {  
    if (amount > 0 && amount <= this.dataBalance) {  
        this.dataBalance -= amount;  
    }  
}  
}  
  
class main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt();  
        sc.nextLine();  
        Customer[] customers = new Customer[n];  
  
        for (int i = 0; i < n; i++) {  
            int customerId = sc.nextInt();  
            sc.nextLine();  
            String customerName = sc.nextLine();
```

```

double initialBalance = sc.nextDouble();
double rechargeAmount = sc.nextDouble();
double usageAmount = sc.nextDouble();

Customer customer = new Customer(customerId, customerName,
initialBalance);

customer.recharge(rechargeAmount);
customer.useData(usageAmount);

customers[i] = customer;
}

for (Customer c : customers) {
    System.out.println("Customer ID: " + c.getCustomerId());
    System.out.println("Customer Name: " + c.getCustomerName());
    System.out.printf("Final Data Balance: %.1f GB\n", c.getDataBalance());
}

sc.close();
}
}

```

Status : Correct

Marks : 10/10

4. Problem Statement

Meera is working as a developer for CityGas Supply Board, which wants to build a household gas billing system.

Each household's gas account has:

A Customer ID (integer) A Customer Name (string) Units Consumed in cubic meters (double)

The gas bill is calculated based on these rules:

For the first 50 units 4 per unit For the next 100 units (51–150) 6 per unit
 For units above 150 8 per unit If the total bill exceeds 2000, a 15% discount is applied on the final bill.

Meera has been asked to implement this system using:

A class with attributes for customer details. A constructor to initialize customer details. Setter methods to update details if needed. Getter methods to retrieve details. Objects of the class to represent customers.

Finally, display each customer's details and final bill amount.

Input Format

The first line of input contains an integer N, representing the number of customers.

For each customer:

- The next line contains the Customer ID (integer).
- The following line contains the Customer Name (string).
- The next line contains the Units Consumed (double).

Output Format

For each customer, print the details in the following format:

Customer ID: <customer_id>

Customer Name: <customer_name>

Final Bill: <final_bill> (The final bill must be rounded to one decimal place.)

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1

1001

Ravi Kumar

30

Output: Customer ID: 1001

Customer Name: Ravi Kumar

Final Bill: 120.0

Answer

```
// You are using Java
import java.util.*;
```

```
class Customer {
    private int customerId;
    private String customerName;
    private double unitsConsumed;

    public Customer(int customerId, String customerName, double
unitsConsumed) {
        this.customerId = customerId;
        this.customerName = customerName;
        this.unitsConsumed = unitsConsumed;
    }

    public void setCustomerId(int customerId) {
        this.customerId = customerId;
    }

    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }

    public void setUnitsConsumed(double unitsConsumed) {
        this.unitsConsumed = unitsConsumed;
    }

    public int getCustomerId() {
        return customerId;
    }

    public String getCustomerName() {
        return customerName;
    }

    public double getUnitsConsumed() {
        return unitsConsumed;
    }

    public double calculateBill() {
        double units = this.unitsConsumed;
        double bill = 0;
```

```

        if (units <= 50) {
            bill = units * 4;
        } else if (units <= 150) {
            bill = (50 * 4) + ((units - 50) * 6);
        } else {
            bill = (50 * 4) + (100 * 6) + ((units - 150) * 8);
        }

        if (bill > 2000) {
            bill *= 0.85;
        }

        return bill;
    }
}

class GasBillingSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        sc.nextLine();
        Customer[] customers = new Customer[n];

        for (int i = 0; i < n; i++) {
            int customerId = sc.nextInt();
            sc.nextLine();
            String customerName = sc.nextLine();
            double units = sc.nextDouble();

            customers[i] = new Customer(customerId, customerName, units);
        }

        for (Customer c : customers) {
            System.out.println("Customer ID: " + c.getCustomerId());
            System.out.println("Customer Name: " + c.getCustomerName());
            System.out.printf("Final Bill: %.1f\n", c.calculateBill());
        }

        sc.close();
    }
}

```

Status : Correct

Marks : 10/10